# MOHAMMED MEHWISH V M

Kerala, India | +91 99616 28750 | michumehwish2004@gmail.com
linkedin.com/in/mohammed-mehwish-202677

## PROFESSIONAL SUMMARY

Mechatronics Engineering graduate with a strong foundation in Robotics and Embedded Systems. Hands-on experience with ESP32, Arduino, Raspberry Pi, C/C++, Python, sensor integration, motor control, and hardware-software interfacing. Knowledge of ROS 2, robot kinematics, OpenCV, communication protocols, and CAD modelling. Strong problem-solving and system integration skills, seeking a Robotics Engineer position focused on developing reliable robotic and autonomous systems.

## SKILLS

- **Robotics:** ROS 2, Robot Kinematics, URDF/Xacro, Gazebo, SLAM, Nav2, Sensor Integration, Actuator Control, Motor Control, Motion Planning
- **Embedded Systems:** ESP32, Arduino, Raspberry Pi, Embedded C, Microcontroller Programming, Hardware–Software Integration
- **Programming:** C, C++, Python
- **Computer Vision & AI:** OpenCV, Machine Learning, Deep Learning, YOLO, Object Detection
- **Electronics & PCB Design:** Circuit Design, PCB Design, GPIO, PWM, ADC, UART, $I^2C$, SPI, Sensor Data Acquisition
- **CAD & Mechanical Design:** Fusion 360, CAD Modelling, Mechanical Design
- **Core Competencies:** System Integration, Problem Solving, Technical Troubleshooting, Team Collaboration

## EDUCATION

**Bachelor of Engineering (B.E.) — Mechatronics Engineering**
Hindusthan College of Engineering and Technology, Coimbatore, Tamil Nadu

## INTERNSHIP EXPERIENCE

**Robotics Intern** **Jul 2026 - Present**
Evolve Robotics, Kochi, Kerala

- Hands-on training in robotics, embedded systems, ROS 2, sensors, and motor control.
- Working with ESP32, Arduino, Raspberry Pi, C/C++, Python, and robotic system integration.
- Developing practical skills in robot kinematics, CAD modelling, and robotics applications.

**Embedded Systems Intern** **Jul 2024 - Aug 2024**
Centre for Skill Enhancement & Employability Development (CSEED), Thrissur, Kerala

- Developed and tested microcontroller-based applications using Embedded C and hardware interfacing.
- Worked with sensors, circuits, data acquisition, and control logic.

## CERTIFICATIONS & PROFESSIONAL TRAINING

- Industrial Automation (PLC, SCADA, HMI, VFD) — Prolific Systems & Technologies Pvt. Ltd. (NSDC Authorised Training Partner), 2024.
- Internship in Embedded Systems — Centre for Skill Enhancement & Employability Development (CSEED), affiliated to NACTET / NCT, 2024.
- Certificate of Participation, Innovation Challenge — TechnoXian World Robotics Championship World Cup 2024, Noida.
- Internship in Introduction to Robotics — D-Rube Labs and Research Pvt. Ltd., Ernakulam, 2023.

## PROJECTS

**Autonomous Mobile Robot**

- Built and simulated a differential-drive autonomous robot using ROS 2 Humble, URDF/Xacro, Gazebo, and Python, with TF and ROS 2 node communication for navigation development.

**Fire Rescue & Surveillance Drone**

- Developed an aerial drone system for fire surveillance and rescue support, combining flight control, real-time monitoring, and sensor integration.

**Helmet Detection using Deep Learning**

- Built a computer-vision model to detect helmet usage in real time using deep learning, supporting automated road-safety compliance monitoring.

## LANGUAGES KNOWN

English | Malayalam | Hindi | Tamil